MILITARY SPECIFICATION

COUPLER, ANTENNA: CU-2064/GRC-193

This specification is approved for use by the Marine Corps, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type antenna coupler.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks for the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	-	Treatment And Painting of Materiel.
MIL-M-13231	-	Marking of Electronic Items.
MIL-P-15024	-	Plate, Tags and Bands For Identification of Equipment.
MIL-E-16400	-	Electronic, Interior Communication and Navigation Equipment Naval Ship and Shore; General Specification.
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing Of.
MIL-A-28988	-	Amplifier, Radio Frequency: AM-6545/GRC-193
MIL-E-52798	-	Enamel, Alkyd, Camouflage

STANDARDS

MILITARY

MIL-STD-108	-	Definition Of And Basic Requirement For Enclosure For Electric And Electronic Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-129 - Marking for Shipment And Storage.
MIL-STD-242 - Electronic Equipment Parts (Selected Standards).
MIL-STD-454 - Standard Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463 - Definition And System Of Units, Electromagnetic Interference Technology.
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipment).
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-781 - Reliability Tests, Exponential Distribution.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-831 - Test Rports, Preparation of
MIL-STD-1281 - Internal Transient Control For Solid State Power Supplies.
MS3114 - Connector, Receptacle, Electric, Serie 1, Solder Type, Jam Nut Mounting Nut, Bayonet Coupling.

HANDBOOKS

MILITARY

MIL-HDBK-472 - Maintainability Prediction

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Prior to submission of any complete production items for acceptance, the contractor shall furnish 10 first articles for determination of conformance to this specification. Examination and tests shall be those specified herein. Approval of the first articles by the procuring activity shall not relieve the contractor of his obligation to supply couplers conforming to this specification. Any changes or deviations of the production items from the first articles shall be subject to the approval of the contracting officer.

3.2 Parts, materials and processes. Parts, materials, and processes approved in MIL-E-16400 shall be used unless otherwise directed by the procuring activity. Parts shall be selected from MIL-STD-242 wherever possible. All nonstandard parts shall be approved by the procuring procuring activity before delivery of couplers. Particular attention shall be paid to the design, application and selection of parts, including

solid state devices, as specified in MIL-E-16400. Parts and solid state units shall be used in such a manner that over-stressing, or operation near maximum rating, shall not be allowed to occur.

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3.3 Design and construction. The coupler shall contain all the necessary circuitry to function as an antenna coupler withe performance as specified herein. No external add-on items shall be required. Provision shall be made to permit automatic tuning operation of the coupler with transmitters having 25 to 400 watts (W) average available for tuning. The equipment shall include automatic protective circuitry in the event of damaging voltage standing wave ratio (vswr) and significant change in antenna impedance. As part of the vehicular AN/MRC-138 Radio Set, the coupler shall be electrically and mechanically compatible and shall operate with the following ancillary equipment:

(a) Amplifier, Radio Frequency AM-6545/GRC-193

(b) Receiver Transmitter RT-1209/URC

(c) Mounting, Electrical Equipment MT-4495/GRC-193

(d) Mounting, Electrical Equipment MT-4497/GRC-193

(e) 15-foot whip antenna and 35-foot whip antenna

(f) Antenna AS-2259

(g) Cable, DC Power CX-( )/GRC-193

(h) Cable, Control CX-( )/GRC-193 (AM-6545 to RT-1209/URC interconnection)

(i) Cable, Control CX-( )/GRC-193 (AM-6545 to CU-2064 interconnection)

The coupler shall meet the requirements specified herein when installed at the base of the associated antenna and interconnected with up to 24 inches of copper conductor. The coupler shall also meet the requirements specified herein when the coupler and its associated antenna are separated from the transmitter with up to 100 feet of interconnecting cable.

3.3.1 Frequency range. The equipment shall be capable of continuous tuning to any frequency in the range of 2.0 to 30.0 megahertz (MHz) with the antenna impedances specified in 3.3.7.

3.3.2 Power sources. The coupler shall operate with any standard direct current (dc) vehicular power supply (+26.5 direct current volts (Vdc) nominal voltage) or

any other source providing +26.5 Vdc nominal voltage. A diode shall be incorporated which will protect the electronic circuitry and internal wiring if the polarity of the power source is reversed from the normal condition where the negative side is grounded.

3.3.3 Input voltage. The coupler shall meet requirements specified herein with input voltages between 22.0 and 30.0 Vdc. This requirement shall be met over the full ambient temperature range of the equipment.

3.3.4 Input power. When operated as specified in 3.3.3, the input power shall not exceed 250 W in any mode of operation.

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3.3.5 Internal transient control. The design of the coupler shall be such as to avoid internal transients which could result in degradation or catastrophic failure of the equipment components. The coupler shall not produce voltage, current or power transients during the course of operation which affect the life of the semiconductor complement or create an unsafe condition. Internal transient control shall be in accordance with MIL-STD-1281.

3.3.6 Radio frequency (rf) power capability. The coupler shall be capable of handling lower sideband, upper sideband (usb), independent sideband, frequency-shift keying, continuous wave (cw) and compatible amplitude modulation (am) transmissions, under continuous operating conditions, with a nominal power up to levels of 400 W peak-envelope-power (pep) and average power over the 2.0- to 30.0-MHZ frequency range. The coupler shall not be damaged by a short or open circuit on the rf output connector with full power applied to the coupler.

3.3.7 Antenna impedance characterics. The antenna impedance characteristics specified in Table I shall be used for the design of the coupler tuning elements. The coupler shall also tune a doublet antenna with the use of a coaxial connector (see 3.3.24). The doublet may have a maximum vswr of 3:1. The coupler shall tune vehicular mounted 35-foot and 15-foot (erect or folded) whip antennas with or without ground staking and the AS-2259 Antenna.

3.3.8 Voltage standing wave ration. The coupler shall automatically match any specified antenna impedance to a 50-ohm transmission line with a vswr of not greater than 1.4:1.

3.3.9 Tune power. The coupler shall automatically tune with a constant forward power carrier level from 25 to 400 W.

3.3.10 Tuning. The coupler shall automatically tune within a time period of not greater than 6 seconds and shall, when tuned, provided an impedance match not exceeding 1.4:1 vswr (see 3.3.8) over 2.0- to 30.0-MHz range. The tune power is specified as 25 to 400 W rf power over the entire frequency range of 2 to 3 MHz. Upon receipt of a retune command, the coupler tuning elements shall proceed to the "home" position. During this portion of the cycle, the associated transmsitter shall be inhibited from transmitting. Upon reaching the "home" position, the transmit inhibit signal shall be released. When the tune rf power is received, the coupler shall automatically tune the antenna, as refined in 3.3.7 to the 50-ohm transmission line within a vswr of 1.4 to 1. When this is completed, normal transmission of signals from the associated transmitter shall be allowed. The equipment shall tune only to the desired operating frequency and shall not tune to a harmonic or other undesired frequencies of the associated transmitter. Tuning shall not be dependent upon frequency coding information from the associated transmitter or other source external to the coupler. When in a tuned condition, surveillance of the transmitter rf shall permit fine tuning under full power to maintain vswr specified in 3.3.13 to correct for any changes in antenna impedance. Simultaneously, protective circuitry surveillance shall protect the equipment from excessive vswr (see 3.3.14). The design requirement is a fully automatic-tuned coupler which will permit continued operational use with 400-W average power applied. In the event of failure of the coupler, the equipment shall provide the necessary control line information to prevent keying of the associated transmitter.

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TABLE I

ANTENNA IMPEDANCE CHARACTERISTICS

FREQ. (MHZ)	15-FOOT WHIP ANTENNA			AS2259 ANTENNA			35-FOOT WHIP ANTENNA		
	EFF(%)	R(OHMS)	X(OHMS)	EFF(%)	R(OHMS	X(OHMS)	EFF(%)	R(OHMS)	X(OHMS)
2	12	2.6	-j950	60	5	-j155	20	2	-j465
3.0	25	3.4	-j650	45	6	-j90	40	3.5	-j280
4.0	35	4.2	-j480	80	50	-j25	65	6.5	-j173
5.0	45	5.0	-j340	55	6	-j63	70	20	-j98
6.0	55	6.0	-j270	75	12	-j10	80	29	-j24
7.0	65	8.0	-j200	65	6	-j65	80	55	+j55
8.0	70	10	-j170	55	3.5	-j30	80	135	+j173
10.0	75	17	-j100	75	16	-j50	80	600	-j57
13.5	85	50	+j0	—	—	—	—	—	—
16	80	100	+j100	—	—	—	80	23	-j99
20	80	400	+j400	—	—	—	80	69	+j32
22	80	700	-j400	—	—	—	80	178	-j29
25	80	300	-j130	—	—	—	80	90	-j121
30	80	90	-j30	—	—	—	75	27	-j21.5

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3.3.11 Removal of rf signal. Given a tuned condition at any frequency from 2 to 30 MHz, upon removal of the associated transmitter's rf signal, it shall be possible to reapply the same rf signal (at the same frequency) without initiating a new cycle.

3.3.12 Efficiency. With the impedance specified in Table I, the equipment shall automatically tune and provide a coupler unit rf input to output efficiency of not less than than specified in Table I.

3.3.13 Surveillance. The couple shall be designed to maintain the vswr to not greater than 1.4:1 with minor changes in the antenna impedance and changes in the operating frequency of ±10 kilohertz (kHz) once the coupler is tuned.

3.3.14 High vswr. The coupler shall be provided with a circuit which automatically prevents damage of the associated transmitter due to any mismatch condition where the vswr is not less than 2:1. This condition shall automatically fault the coupler and at the same time disrupt the transmitter key disable line. The circuit shall be enabled only in the tuned mode of operation.

3.3.15 Voltage standing wave ratio monitoring. A directional coupler with the capability of measuring the forward and reflected power shall be incorporated into the coupler rf input circuit with the dc leads available at the control line receptacle for use with external meters. The sensitivity and source impedance shall be compatible with the metering requirements of the amplifier specified in 3.3.

3.3.16 Fault circuitry. To prevent damage to the coupler and the associated equipment, the fault circuitry shall provide protection whenever tuning or homing does not occur within 10 seconds, and at which time it shall produce a fault signal for external use.

3.3.17 Servo amplifiers. No balancing of servo amplifiers shall be necessary at any time to maintain performance. If more than one servo amplifier is used, they shall be directly interchangeable. The servo amplifiers shall have sufficient interlocking to prevent shorting of the power supply and damage to the servo amplifier.

3.3.18 Servo motors. High reliability dc torque type motors shall be used.

3.3.19 High power rf switching. Any hot switching of high-power rf circuits shall be accomplished by means of hermetically sealed relays.

3.3.20 Amplifier to coupler interface. The coupler shall provide the following control signals for use with the associated amplifier:

(a) Transmit inhibit - open for normal operation, ground when the coupler is not ready to accept rf power input, or primary dc power is switched OFF.

(b) Fault - ground when a coupler fault is present, open normally.

(c) Direct current analog of forward power.

(d) Direct current analog of reflected power.

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(e) Ready - ground when coupler has tuned and is ready for normal transmit operation.

The coupler shall accept and operate with the following control signals:

(a) Primary Power +26.5 Vdc (nominal primary power supply)

(b) System dc ground

(c) Keyline - ground for transmit mode of operation

(d) Tune Start - momentary ground pulse whenever a frequency digit is changed.

(e) Shield ground - ground connection to the shield of the interconnection cable.

(f) Blower +26.5 Vdc

(g) Coupler on - 26 Vdc (nominal) through 3.3

(h) Fan Test - +26.5 Vdc energizes external blower

(i) Power amplifier (pa) Meter ground return

3.3.21 Receive only. The coupler shall have relays to bypass the variable tuning network. When no primary power is applied or when the coupler is in the untuned condition, the elements shall be bypassed. This condition shall allow receive signals from the antenna to be conducted through the coupler to the rf input connector of the coupler with a minimum of attenuation.

3.3.22 System interface. The amplifier-receiver/exciter and amplifier-coupler control interface lines shall be in accordance with Tables II and III respectively. All interface lines (except grounds) shall be radio frequency interference filtered before exiting the amplifier shield enclosure.

3.3.23 Continuous duty. The coupler shall operate as specified herein under continuous locked key duty, without the requirement for special external cooling facilifies. An internal blower shall be energized continuously by application of primary power only. It shall be deenergized in the receive only mode of operation.

3.3.24 External controls and receptacles. The following external controls and receptacles shall be provided:

(a) Control cable receptacle, MS3114E-14-15P

(b) Radio frequency output connector for the whip antennas and AS-2259 Antenna - high voltage threaded type.

(c) 50 ohm rf output connector - (N)

(d) Radio frequency input connector (N)

(e) Two ground studs with wing nuts (one for grounding coupler to the vehicle, one for connection to ground rod when used)

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TABLE II - AMPLIFIER TO RECEIVER-EXCITER INTERFACE

RT-1209/URC Receiver/Exciter				AM-6545/GRC Amplifier	
Line Conditions/Status	J2N2026@000 Pin Connections	Function	Signal Source Pin Connection	Line Condition/Status	Operating Limits
0-Logic-Hard and Thru Diode and Switch Contacts /200 ohms Maximum 1-Logic-Open	L M N P R S	Frequency Code, 1 MHz (8) Frequency Code, 1 MHz (4) Frequency Code, 1 MHz (2) Frequency Code, 1 MHz (1) Frequency Code, 10 MHz (2) Frequency Code, 10 MHz (1)	A B C D E F	Max "0" Current - 1MA 4 VDC pull voltage thru 15000 ohms Under Logic "1" Condition	Frequency Decoder Circuits Shall Operate Properly with the Following Limit Inputs: 0-GND thru 500 ohms Max 1-50,000 ohms Min to GND
Sidetone-2ma " x sink current $\overline{\text{Sidetone}}$ +8 VDC thru 10Kohm Pullup	F	Sidetone	H	GND thru Collector. Emmiter Junction of Saturated Negative Positive Negative Transistor and 560 ohms When TX-RF Output Present. Open Collector-Emitter Junction in RX and TX (No signal)	
TIP-2ma max sink current $\overline{\text{TIP}}$ +8 VDC thru. 10 K Pullup	K	Tune in Progress (TIP)	J	GND Thru C-E Junction of Xsistor and 560 ohms during Auto Tune Cycle Open C-E Junct. in Normal RX-TX Operation	
Keyed thru Diode and Collector Emitter Junction of Transistor Keyed Open	F	Keyline	K	Keyed 5 ma Max Sink Current Keyed 20.5 VDC thru 10,000 ohms	The Keying Circuits Shall Operate Properly with the following limit Inputs: Keyed-GND thru 15,000 ohms Max Keyed-50,000 ohms Min to GND
	U	+26.5 VDC (nom)	L	Direct Connections to +26.5 Primary Supply Line	+22 to 30 VDC
Direct Connection to Chassis GND	A	GND	M	Direct Connection to Chassis GND Inside PA	
Off-Open, 200,000 ohms Min ON-GND thru C-E Junction of Transistor	T	PA On-Off Switching	N	Off +26.5 VDC thru 10,000 ohms ON-Current 8 MA Max	The On-Off Circuits Shall Operate Properly with the following Limit Inputs: On-GND thru 400 ohms Max Off 50,000 ohms Min to GND
Tune Start-GND Pulse thru Collector-Emitter Junction GND Pulse Width 30 ms Min Series Resistance 200 ohms Max $\overline{\text{Tune Start}}$ 26 VDC Nom thru 12 K ohms	C	Tune Start	P	Tune Start-13 ma Max Sink Current Tune Start 5.0 VDC thru 10,000 ohms and Diode (Amplifier) or +26.5 VDC thru 5700 ohms and Diode (with antenna coupler)	The Tune Start Circuits Shall Operate Properly with the following Limit Inputs: Tune Start-2 Milliseconds Min GND Pulse Thru 300 ohms Max Tune Start 50,000 Min to GND
Load Impedance 50,000 ohms Min	J	ALC/Transmitter Control	R	Positive Going Analog of RF Power Output Source Impedance Less than 2,000 ohms adjustable Level.	
	M	Fault		Fault (Amplifier or Antenna Coupler) GND thru Collector-Emitter Junction of Negative Positive Negative Silicon Transistor and Blocking Diode Fault-Open Diode Isolated From 26 V.	
Direct Connection to Chassis GND	B	Shield GND	U	Direct Connection to Chassis GND Inside PA	
Ant CPLR "On" 26.5 VDC Ant CPLR "Off" Open	D	Ant CPLR On-Off Switching	T	Direct Connection to Antenna Coupler (See Table II)	+22 to +30 VDC Max for Rx-Only Switching

NOTE: $\overline{X}$ - Negation of A

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Table III- Amplifier to Antenna-Coupler Interface

CU-2064/GRC Antenna Coupler Line Condition/Status	MS3114M14-15P Pin Connections	Line Function	MW20F(M)D00 Pin Connections	AM-6545/GRC Amplifier Condition/Status	Operating Limits
CPlr on 10mA Max current sink CPlr on (Normal Rx-Tx)-open, diode isolated	A	Antenna Coupler ON/Off	A	Direct connection to pin "T" of the amplifier-receiver exciter connector (See Table I)	
Fault-GND through collector emitter junction of saturated silicon NPN transistor Fault - Open collector emitter junction	B	Fault	B	Fault- 5mA Max sink current Fault-20.5V through 10K and diode	The fault sensing circuits shall operate properly with the following limit inputs: Fault - GND through 1000 ohms Max Fault - 5000 ohms min to GND
Ready -GND through collector-emitter junction of saturated NPN silicon transistor Ready-Open collector emitter junction	C	Ready	C	Ready-2mA Max sink current READY-10 Vdc (NOM) through 1000 ohms	The ready sensing circuits shall operate properly with the following limit inputs: Ready - GND through 1000 ohms Max Ready-5000 ohms min to GND
Tx inhibit-GND through collector emitter of saturated silicon NPN transistor Rx inhibit-open collector	D	Transmit Inhibit	D	Interlock-10mA max sink current Interlock-+26.5 Vdc (NOM) Max through 2700 ohms	The key interlocking circuit shall operate with the following limit inputs: Interlock-GND through 50 ohms Max
Tx inhibit cont'd emitter junction (normal tune)					Interlock-1800 ohms Min to GND
Direct connection to chassis GND	E	GRD	E	Direct connection to chassis GND inside PA	
Tune start - 10mA Max current sink Tune start -+26.5Vdc through 5700 ohms and diode	F	Tune Start	F	Direction connection to Pin "F" of the amplifier-receiver/ exciter connector (See Table I)	
Positive going DC analog of RF power, forward	H	Forward Power	H	Loaded by 10000ohms and the 500 micro-ampere meter move-ment	
Tx-10 mA sink current Tx-10 Vdc through 1800 ohms and diode	K	Keyline	K	Tx-Gnd through collector emitter junction of saturated NPN silicon transistor Tx open	
DC-AC inverter load (One or two blowers operating) 4.0 Max current drain	L	Blower +26.5 Vdc	L	Switched on/off by the amplifier primary power relay. Off-open line to +26.5 Vdc(NOM) primary line, system off or Rx-only mode. On +26.5 Vdc (NOM) in system on normal Rx-Tx mode	+22 to 30 Vdc
3.6A Max current drain in tune cycle	M	+26.5 Vdc primary power	M	Direct connection to +26.5 Vdc (NOM) primary supply line	+22 to 30 Vdc
Direct connection to chassis GND	Z	Shield GND	P	Direct connection to chassis GND inside PA	
Fan test -50mA Max current sink Fan test-GRD through relay coil resistance of 700 ohms Max	N	Blower test	N		

Note: $\overline{A}$ negation of A

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The rf output connectors (d and e above) shall be mechanically interlocked in such a way that it is impossible to make electrical connections to both connectors simultaneously.

3.3.25 Hardware. All screws and nuts used shall be austenitic corrosion-resisting steel except where high rf fields require the use of brass or copper. Corrosion-resistant steel crossed-recessed drive screws and bolts shall be employed in all screw and bolt applications throughout the coupler in accordance with MIL-E-16400 (head styles and drive). Captive type bolts shall be used on the cover of the case and electronic plug-in assemblies.

3.3.26 Fasteners to vehicular mounts. The coupler shall be fastened to the Mounting, Electricl Equipment MT-4495/GRC-193 with screw-clamp fasteners.

3.3.27 Enclosure. The enclosure for the coupler unit shall be of the aluminum, head-exchanger type and shall be watertight as specified in MIL-STD-108. Two handles shall be mounted on the end of the coupler to protect protrusions and to facilitate hand-carrying.

3.3.27.1 Gasket. Particular attention shall be made in the selection of a gasket which will conform with enclosure specified in 3.3.27. The gasket shall be a permanent type, reusable for the life of the coupler, regardless of the number of openings for repair.

3.3.27.2 Case and cover interface. A means shall be provided to prevent galvanic action between the case and the cover connectors and bolts of the coupler unit in accordance with MIL-E-16400, Table I.

3.3.28 Warm-up time. The coupler shall require no warm-up time.

3.3.29 Electrical overload protections. Electrical overload protection for the coupler shall be in accordance with Requirement 8 of MIL-STD-454.

3.3.30 Accessibility. All modules and components shall be easily accessible for troubleshooting and replacement. Accessible module test points shall be provided for use with standard test equipment, authorized at direct and general support, to locate and replace faulty modules.

3.3.31 Human engineering. Human engineering shall conform to Requirement 62 of MIL-STD-454.

3.3.32 Interchangeability. Provisions for interchangeability are required and shall be in accordance with Requirement 7 of MIL-STD-454.

3.3.33 Safety. Personnel safety hazards shall be kept to a minimum in accordance with Requirement 1 of MIL-STD-454.

3.4 Reliability. Reliability of operation shall be considered of prime importance in the design and manufacture of the coupler. The contractor shall employ all methods possible in the process of design, development and manufacture which will assure quality and maximum reliability consistent with the state-of-the-art. In the functional application of parts, adequate factors of safety shall be provided by suitable derating from the part specification values in order to assure

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high equipment reliability under all applicable service conditions. The design shall include all possible features which will result in reliable and stable operation and hence reduce the requirements for maintenance by highly skilled personnel.

3.4.1 Reliability program. The contractor shall conduct a reliability program which will provide for, but shall not be limited to, the following reliability tasks as described in MIL-STD-785:

(a) Reliability organization

(b) Management and control

(c) Subcontractor and supplier reliability programs

(d) Program reviews

(e) Design techniques

(f) Reliability analysis

(g) Parts reliability

(h) Reliability critical items

(i) Effects of storage, shelf-life, packaging, transportation, handling and maintenance

(j) Reliability demonstration

(k) Failure data collection, analysis and corrective action

(l) Status reports

3.4.2 Numerical reliability requirement. The reliability shall be a 1,000 hour mean-time-between-failure (MTBF). This shall be achieved under the following operating conditions:

(a) Ambient operating temperature: 50° C

(b) Vehicle mounted ground exposed environment

(c) Duty cycle: 9 to 1 receive to transmit ratio

3.4.3 Tuning life. The automatic tuning system shall perform a total of 6,000 tune cycles without failure or degradation requiring replacement of components or adjustment.

3.5 Maintainability. Design criteria for the maintainability of the coupler and support equipment shall be in accordance with MIL-STD-470, MIL-STD-471 and MIL-HDBK-472. The mean-time-to-repair (MTTR) for the replacement of defective modules shall be no more than 15 minutes. The MTTR for the equipment, modules, and subassembles at the depot levels shall be no more than 1 hour.

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3.6 Performance. The coupler shall be capable of the following performance.

3.6.1 Operation. The coupler shall operate in wheeled or tracked vehicles, either stationary or in motion, powered by the normal primary dc power source of the vehicle.

3.6.2 Compatibility. The coupler shall be operationally compatible with the Amplifier, Radio Frequency: AM-6545/GRC-193 confirming to MIL-A-28988.

3.6.3 Modes of operation. The coupler shall handle voice communications using single sideband (ssb) and compatible am (usb plus carrier) and cw morse communications at frequencies from 2 to 30 MHz.

3.6.4 Standard conditions. The following standard conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input Voltage	26.5 $\pm$0.3

3.6.5 Service conditions. The coupler shall be capable of operating under the following service conditions:

(a)	Ambient temperature range	-40°C to +55°C
(b)	Ambient humidity range	0 to 100 percent relative
(c)	Vibration	Method 514, Procedure VIII. Schedule A, Curve V, 5-55 Hz 2.5g or 0.15" double amplitude whichever is the lesser in MIL-STD-810.
(d)	Shock	Method 516, Procedures I, II and V of MIL-STD-810
(e)	Submersion	Method 512, Procedure I of MIL-STD-810
(f)	High temperature	Method 501, Procedure II of MIL-STD-810, except that at Step 4 the temperature shall be +55°C.
(g)	Low temperature	Method 502, Procedure I of MIL-STD-810
(h)	Humidity	Method 507, Procedure IV of MIL-STD-810.

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(i)	Altitude	Method 500, Procedure I of MIL-STD-810
(j)	Salt fog	Method 509, Procedure I of MIL-STD-810
(k)	Dust	Method 510, Procedure I of MIL-STD-810
(l)	Fungus	Method 508, Procedure I of MIL-STD-810
(m)	Rain	Method 506, Procedure I of MIL-STD-810

3.6.6 Storage condition. The coupler shall be capable of being stored under the following conditions without degradation:

(a)	Ambient temperature range	-62°C to +65°C
(b)	Altitude	Sea level to 70,000 feet

3.7 Electromagnetic-interference suppression. The coupler shall meet the electromagnetic interference control limits of MIL-STD-461 using the test methods of MIL-STD-462. Terms and definitions shall be accordance with MIL-STD-463.

3.8 Dimensions. The maximum overall dimensions of the coupler, excluding external projections such as handles, connectors and cover holddown protrusions shall be as follows:

14.80 inches long
11.40 inches wide
8.05 inches high

3.9 Weight. The weight of the coupler shall not exceed 42 pounds.

3.10 Treatment and painting. Treatment and painting of the coupler shall be in accordance with MIL-T-704, type A; the finish coat shall conform to MIL-E-52798 (forest green).

3.11 Marking and identifications. Marking shall conform to MIL-M-13231, and the following. Panel markings for switch positions shall be accurately aligned with detent positions.

3.11.1 Nameplate. The coupler shall contain name plates in accordance with MIL-E-16400 and MIL-P-15024.

3.12 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3).

(b) Production (see 4.4).

(c) Sample (see 4.5).

4.3 First article inspection. The ten first articles shall be subjected to tests indicated for first article in Table IV. The Government representative present will select which coupler is to be subjected to each test.

4.3.1 First article test reports. After completion of the first article tests, the contractor shall prepare a test report.

4.4 Production inspection. All couplers shall be subjected to the production tests indicated in Table IV.

4.5 Sample inspection. During the production phase, sample tests shall be conducted as indicated in Table IV. The following quantities shall be selected by the Government inspector for sample tests:

Quantity of Couplers Offered for Acceptance	Quantity to be tested
First 10	1
Next 90	1
Next 100	1
	One for each additional 100 or fraction thereof.

4.6 Test plans. Test plans describing in detail first article, production and sample testing shall be prepared by the contractor in accordance with the requirements of MIL-STD-831 and submitted to the contracting officer for approval prior to initiation of testing. Approval of the test plan by the contracting officer or his representative is required prior to commencement of the testing. The test plans shall include all test procedures deemed necessary and practicable by the contractor to demonstrate compliance with this specification.

4.7 Tests.

4.7.1 Service condition tests. When a test requires measurements to be made before, during or after the test, these measurements shall be accomplished as specified in the test plans of 4.6. For those service condition tests where electrical measurements are not specified, tests specified in 4.7.4.1 shall be made, for the 15-foot whip antenna impedances, prior to performing the service condition test and at the conclusion of the service condition test. For those service condition tests where electrical measurements are not specified and the equipment is required to meet specified performance during testing, as a minimum, the measurements above shall also be made during the test.

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TABLE IV - TESTS

TEST	REQT. PARA.	TEST PARA.	FIRST ARTICLE	PRODUCTION	SAMPLE
Mechanical Inspection	3.3	4.8.2	x		x
Frequency Range and vswr	3.3.1	4.7.4.1	x	x	x
Radio Frequency Power Capability	3.3.6	4.7.4.2	x	x	x
Automatic Tuning and Retuning	3.3.10	4.7.4.3	x	x	x
Tuning Life	3.4.3	4.7.25	x		x
Efficiency	3.3.12	4.7.4.4	x	x	x
Reliability	3.4	4.7.2	x		
Continuous Duty	3.3.23	4.7.4.5	x		x
Maintainability	3.5	4.7.3	x		
Surveillance	3.3.13	4.7.4.6	x	x	x
Vibration	3.6.5	4.7.1.4	x		x
Shock	3.6.5	4.7.1.5	x		x
Submersion	3.6.5	4.7.1.7	x		x
High Temperature	3.6.5	4.7.1.1.2	x		x
Low Temperature	3.6.5	4.7.1.1.1	x		x
Humidity	3.6.5	4.7.1.2	x		
Altitude	3.6.5	4.7.1.3	x		x
Salt Fog	3.6.5	4.7.1.8	x		
Dust	3.6.5	4.7.1.6	x		
Fungus	3.6.5	4.7.1.9	x		
Rain	3.6.5	4.7.1.10	x		
System Operational Inspection	3.3.20	4.8.1	x		x

NOTE: SEQUENCE OF SERVICE CONDITION TESTS SHALL BE IN ACCORDANCE WITH TABLE I OF MIL-STD-810 FOR GROUP II j EQUIPMENT

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4.7.1.1 Temperature. During the temperature tests, the coupler shall be operated with 400-W average rf input except at the temperature extremes where all modes shall be measured.

4.7.1.1.1 Low temperature. The coupler shall be subjected to low temperature test in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -65°C which shall be maintained until temperature stabilization of the test item is reqeached. The low operating temperature (Step 4) shall be -40°C. The chamber air circulating fan shall be baffled to protect the coupler from direct air flow. Make the measurements listed in 4.7.1 prior to or at Step 1 and at Step 5.

4.7.1.1.2 High temperature. The coupler shall be subjected to high temperature test in accordance with Method 501, Procedure II, of MIL-STD-810, except at Step 4 the temperature shall be +55°C. The high operating temperature for the coupler shall be +55°C. Make the measurements listed in 4.7.1 prior to, or at, Step 1, and at Steps 8 and 10.

4.7.1.2 Humidity. The coupler shall be subjected to humidity test in accordance with Method 507, Procedure IV, of MIL-STD-810. The coupler shall be operated in all modes where measurements are required. Make the measurements listed in 4.7.1 as minimum, at Step 4, Step 5, Step 6, and Step 8.

4.7.1.3 Altitude. The coupler shall be subjected to altitude test in accordance with Method 500, Procedure I of MIL-STD-810. Make measurements as listed in 4.7.1.

4.7.1.4 Vibration. The coupler shall be subjected to vibration test in accordance with Method 514, Procedure VIII, Schedule A, Curve V, 5 to 55 Hz, 2.56 or 0.15 inch double amplitude whichever is the lesser in MIL-STD-810. The coupler shall be operating during this test.

4.7.1.5 Shock.

4.7.1.5.1 Bench handling. The coupler shall be subjected to the test of Method 516, Procedure V, of MIl-STD-810. Make measurements as listed in 4.7.1.

4.7.1.5.2 Drop. The coupler, packaged for shipment, shall be subjected to drop test in accordance with Method 516, Procedure II, of MIL-STD-810. The coupler shall be inspected for damage after completion of test. Make masurements listed in 4.7.1.

4.7.1.5.3 Basic design. The coupler shall be subjected to the test of Method 516, Procedure I, of MIL-STD-810. The shock pulse shape shall be in accordance with figure 516.1, of amplitude a and time duration c. The coupler shall be mounted in its vehicle mounted tray for this test.

4.7.1.6 Dust. The coupler shall be subjected to dust test in accordance with Method 510, Procedure I, of MIL-STD-810. Make measurements as listed in 4.7.1.

4.7.1.7 Submersion. The coupler shall be subjected to submersion test in accordance with Method 512, Procedure I, of MIL-STD-180. No electrical measurements are required.

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4.7.1.8 Salt fog. The coupler shall be subjected to salt fog test in accordance with Method 509, Procedure I, of MIL-STD-810. Operation of the coupler shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the coupler surfaces which can be attributed to flaky, pitted, blistered or otherwise loosened finish of metal surfaces.

4.7.1.9 Fungus. The coupler shall be subjected to fungus test in accordance with Method 508, Procedure I, of MIL-STD-810. Operation of the coupler shall not be required. At the conclusion of the exposure period, examination shall reveal no evidence of viable fungus or corrosion on any surfaces.

4.7.1.10 Rain. The coupler shall be subjected to rain test in accordance with Method 506, Procedure I, of MIL-STD-810. The coupler shall be operating during the last 10 minutes of the 30-minute rain.

4.7.2 Reliability.

4.7.2.1 Reliability acceptance test. The contractor shall conduct a reliability acceptance test in accordance with Test Plan III of MIL-STD-781. The specified MTBF shall be 1,000 hours. The minimum number of couplers to be tested shall be 4. Minimum test time per coupler shall be in accordance with MIL-STD-781. The reliability testing shall be conducted in the sequence specified in MIL-STD-781.

4.7.2.2 Reliability test conditions. Reliability test described above shall be conducted under the following conditions. Test Level C of MIL-STD-781 shall be the test environment. A 9 to 1 receive to transmit duty cycle shall be used. The coupler shall be rechanneled and retuned at the beginning of each transmit period. The frequencies to be used are those of 4.7.1. The frequency shall be changed once every 24 hours. Proper instrumentation shall be provided to insure recognition of failure. A failure is defined as the inability of a coupler to perform its required function. This includes catastrophic failure as well as any event which causes departure from performance specified herein or the contractor's approved test procedure.

4.7.2.3 Rework and retest provisions. Rework and retest provisions shall apply only to reliability acceptance test. Once a reject decision has been reached, no rerun can be made of that test on the same or a new sample from the same lot unless positive action consisting of design change, modification or rework actually relating to observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time.

4.7.2.4 Burn-in (debugging) period. Every coupler shall be subjected to a 24-hour burn-in period. The last 12 hours of burn-in shall be failure free. The burn-in shall be conducted at test level C of MIL-STD-781 and the coupler shall be exercised and monitored in the same manner as during reliability testing. The failures shall be recorded and analyzed but shall not be considered relevant for computation of the coupler MTBF.

4.7.2.5 Tuning life. During the 6,000 tune-ups, various frequencies and dummy loads simulating different antennas shall be used in conjunction with the AM-6545/GRC-193 Amplifier. No more than 25 retunes requiring a second cycle shall be allowed. No retunes in excess of two attempts shall be permitted, and as such, constitute failure of this life requirement. Full power output shall be recorded for each tune cycle.

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4.7.3 Maintainability. The required MTTR shall be demonstrated in accordance with MIL-STD-471. The requirements of 3.5 shall be met.

4.7.4 Electrical.

4.7.4.1 Frequency range and vswr. The frequencies shown below and their corresponding load impedances shown in Table I for each antenna shall be used as a minimum to demonstrate that the coupler will tune to any frequency within the specified range. This test may be performed using a reduced level of power. Successful tuning will be indicated by a vswr of not more than 1.4 to 1 and a tuning time of not more than 6 seconds (see 3.3.1, 3.3.8, and 3.3.10).

Test Frequencies (MHz)

15-Foot Whip Antenna	AS2259 Antenna	35-foot Whip Antenna
2.0	2.0	2.0
4.0	4.0	4.0
8.0	6.0	8.0
13.5	8.0	10.0
16.0	10.0	16.0
22.0		22.0
30.0		30.0

4.7.4.2 Radio frequency power capability. Four hundred (400) W average rf power shall be applied to the coupler at each frequency show below. Corresponding load impedances for the frequences from Table I, 15-foot whip antenna, shall be utilized. During each period of continuous application of rf power, the coupler shall be examined for operation specified in 3.3.6.

Test Frequencies (MHz)

2.0
8.0
30.0

4.7.4.3 Automatic tuning and retuning. The coupler shall conform to the requirements of 3.3.10.

4.7.4.4 Efficiency. First article and sample coupler efficiency tests, throughout the 2.0- to 20.0-MHz frequency range and using the load impedances specified in Tabel I, shall be conducted. A report shall be prepared showing the coupler efficiency. For production testing, the test frequencies should be 2.0, 8.0 and 30.0 MHz.

4.7.4.5 Continuous duty. The coupler shall be subjected to a continuous operation test with 400 W average rf input power at 55°C. The test frequency shall be 2.0, 4.0, 6.0, 13.5 and 30.0 MHz with the corresponding load impedances show in Table I for the 15-foot whip antenna and for 2 hours for each frequency. Heat sensing elements shall be appropriately located on the coupler to detect any abnormal temperature rise. Voltage standing wave ratio readings shall be made periodically throughout the test to determine that the ratio does not exceed 1.4 to 1. At the end of each 2-hour period, perior to reducing the temperature, the coupler shall be operated as specified in 4.7.4.1 for the 15-foot whip antenna impedances only.

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4.7.4.6 Surveillance. Using the 15-foot whip antenna in impedances at 2.0, 6.0, 13.5 and 29.9 MHz, the antenna impedance shall be varied, the carrier frequency shall be adjusted up 10 kHz and the coupler response shall be observed. A vswr of not greater than 1.4:1 shall be maintained. If the coupler should fine tune the antenna impedance, it shall do so without faulting.

4.8 Inspection.

4.8.1 System operational inspection. The coupler shall be used in a complete vehicular system consisting of items in 3.3 and a suitable antenna to communicate with another similar system. The coupler shall be checked for proper operation in all operating modes.

4.8.2 Mechanical inspection. The coupler shall be given a thorough mechanical and visual inspection to determine that the quality of all materials and workmanship is in compliance with the requirements specified herein and in MIL-E-16400. Particular attention shall be given to the following:

(a) Completeness

(b) Nameplates, identification markings and labels

(c) Ease of operation of connector mating, thumb screws, controls and switches and any other part intended for adjustment, movement or operation.

(d) Finishes

(e) Welded joints

(f) Check of solder joints

(g) The fit of components in their respective positions

(h) Check of mounting means

(i) Check of safety features

(j) Fastening and securing devices or parts

(k) Accessibility of components and parts for servicing

(l) Cable run between components including plugs and receptacles

(m) Fit of sealing gasket on cover and wear on gasket

(n) Check of all connector fittings

(o) Overall dimensions

(p) Weight

(q) Other visual defects

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5. PACKAGING

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. This coupler is an automatic tuning unit intended for vehicular use with radio transmitters with nominal rf output power ratings up to 400 W pep and average, whip antennas as short as 15 feet and doublets over the frequency range of 2 to 30 MHz, and the AS2259 Antenna over the frequency range of 2 to 10 MHz. The coupler, as a part of the AN/GRC-193 fixed, and the AN/MRC-138, vehicular system Radio Sets will be used to establish and maintain medium and long range high frequency communications. The equipment will be used in all types of weather and terrain including amphibious and airborne landing operations.

6.2 Order data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following.

6.2.1 Acquisition

(a) Title, number and date of this specification

(b) Levels of preservation, packaging and packing (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisiton which incorporates a DD Form 1423, Contract Data Requirements Lists (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirements	Applicable DID No.	Options
3.4.1	Reliability Program Plan	DI-R-7079	
4.3	First Article Inspection Procedures	DI-T-4901	
4.3.1	First Article Inspection Report	DI-T-4901	
4.7.2.1	Reliability Test Plan	DI-R-7033	
4.7.2.1	Reliability Test and Demonstration Report	DI-R-7034	

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article instructions. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. Selected first article tests will be witnessed by a representative of the procuring activity. Tests not witnessed by a representative of the procuring activity shall be witnessed and certified by the appropriate Defense Contract Administration Services District representative. Approval of the first article shall be by the contracting officer.

6.4 Marginal Notations. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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